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Course Name: Statistics with R Programming

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LAB OBSERAVTION

1) Create numeric, character, and logical vectors and display type and content.

Answer)

Code:

```
# Create vectors
numeric_vector <- c(10, 20, 30, 40, 50)
character_vector <- c("Apple", "Mango", "Orange")
logical_vector <- c(TRUE, FALSE, TRUE, FALSE)
# Display vectors
print("Numeric Vector:")
print(numeric_vector)
print(typeof(numeric_vector))
print("Character Vector:")
print(character_vector)
print(typeof(character_vector))
print("Logical Vector:")
print(logical_vector)
```

print(typeof(logical_vector))

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-1.R")
[1] "Numeric Vector:"
[1] 10 20 30 40 50
[1] "double"
[1] "Character Vector:"
[1] "Apple"  "Mango"  "Orange"
[1] "character"
[1] "Logical Vector:"
[1]  TRUE FALSE  TRUE FALSE
[1] "logical"
>
```

2) Create labeled matrices (5×4, 3×3, 2×2) filled by row/column.

Answer)

Code:

```
# Create a 5x4 matrix filled by rows
m1 <- matrix(1:20, nrow = 5, ncol = 4, byrow = TRUE)
rownames(m1) <- c("R1", "R2", "R3", "R4", "R5")
colnames(m1) <- c("C1", "C2", "C3", "C4")
print("5x4 Matrix:")
print(m1)
# Create a 3x3 matrix filled by columns
m2 <- matrix(1:9, nrow = 3, ncol = 3, byrow = FALSE)
rownames(m2) <- c("A", "B", "C")
colnames(m2) <- c("X", "Y", "Z")
print("3x3 Matrix:")
print(m2)
# Create a 2x2 matrix
m3 <- matrix(1:4, nrow = 2, ncol = 2, byrow = TRUE)
rownames(m3) <- c("First", "Second")
colnames(m3) <- c("Left", "Right")
print("2x2 Matrix:")
print(m3)
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-2 Create Labeled Matrices (5x4, 3x3, 2x2).R")
[1] "5x4 Matrix:"
   C1 C2 C3 C4
R1  1  2  3  4
R2  5  6  7  8
R3  9 10 11 12
R4 13 14 15 16
R5 17 18 19 20
[1] "3x3 Matrix:"
  X Y Z
A 1 4 7
B 2 5 8
C 3 6 9
[1] "2x2 Matrix:"
       Left Right
First     1     2
Second    3     4
>
```

3) Write an R program to create and display a 3D array with specified rows, columns, and tables.

Answer)

Code:

```
# Create a 3D array

arr <- array(

  1:24,

  dim = c(3, 4, 2)

)

print("Three-Dimensional Array:")

print(arr)

print("Dimensions:")

print(dim(arr))
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-3 Create and Display a Three-Dimensional Array.
R")
[1] "Three-Dimensional Array:"
, , 1

     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8   11
[3,]    3    6    9   12

, , 2

     [,1] [,2] [,3] [,4]
[1,]   13   16   19   22
[2,]   14   17   20   23
[3,]   15   18   21   24

[1] "Dimensions:"
[1] 3 4 2
>
```

4) Create arrays from vectors with dimension names, print specific elements.

Answer)

Code:

```
# Create a vector
values <- 1:12
# Create an array with dimension names
arr <- array(
  values,
  dim = c(2, 3, 2),
  dimnames = list(
    Rows = c("R1", "R2"),
    Columns = c("C1", "C2", "C3"),
    Tables = c("T1", "T2")
  )
)
print("Array:")
print(arr)
# Access specific elements
print("Element at R1, C2, T1:")
print(arr["R1", "C2", "T1"])
print("Element at R2, C3, T2:")
print(arr["R2", "C3", "T2"])
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-4 Create Arrays with Dimension Names and Access.R")
[1] "Array:"
, , Tables = T1

    Columns
Rows C1 C2 C3
  R1  1  3  5
  R2  2  4  6

, , Tables = T2

    Columns
Rows C1 C2 C3
  R1  7  9 11
  R2  8 10 12

[1] "Element at R1, C2, T1:"
[1] 3
[1] "Element at R2, C3, T2:"
[1] 12
>
```

5) Create and manipulate factor variables (e.g., women's dataset heights, random LETTERS sample).

Answer)

Code:

```
# Create a vector of heights

heights <- c(150, 160, 155, 170, 165, 160, 155)

# Convert heights into a factor

height_factor <- factor(heights)

print("Height Factor:")

print(height_factor)

print("Height Factor Levels:")

print(levels(height_factor))

# Set seed for reproducibility

set.seed(10)

# Generate a random sample of letters

letters_sample <- sample(LETTERS, 10)

print("Random Letters:")

print(letters_sample)

# Convert letters into a factor

letter_factor <- factor(letters_sample)

print("Letter Factor:")

print(letter_factor)

print("Letter Factor Levels:")

print(levels(letter_factor))
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-5 Create and Manipulate Factor Variables.R")
[1] "Height Factor:"
[1] 150 160 155 170 165 160 155
Levels: 150 155 160 165 170
[1] "Height Factor Levels:"
[1] "150" "155" "160" "165" "170"
[1] "Random Letters:"
 [1] "K" "I" "J" "P" "L" "H" "G" "S" "O" "R"
[1] "Letter Factor:"
 [1] K I J P L H G S O R
Levels: G H I J K L O P R S
[1] "Letter Factor Levels:"
 [1] "G" "H" "I" "J" "K" "L" "O" "P" "R" "S"
>
```

6) Create an R list containing vectors, matrices, and functions; display contents.

Answer)

Code:

```
# Create a vector
v <- c(10, 20, 30, 40)
# Create a matrix
m <- matrix(1:6, nrow = 2, ncol = 3)
# Create a function
square <- function(x) {
  return(x * x)
}
# Create a list
my_list <- list(
  Numbers = v,
  Matrix = m,
  SquareFunction = square
)
# Display list contents
print("Complete List:")
print(my_list)
print("Vector:")
print(my_list$Numbers)
print("Matrix:")
print(my_list$Matrix)
print("Square of 5:")
print(my_list$SquareFunction(5))
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-6 Create an R List
Containing Vectors, Matrices, and Functions.R")
[1] "Complete List:"
$Numbers
[1] 10 20 30 40

$Matrix
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

$SquareFunction
function(x) {
  return(x * x)
}

[1] "Vector:"
[1] 10 20 30 40
[1] "Matrix:"
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
[1] "Square of 5:"
[1] 25
>
```

7) Write R programs for basic tasks: Factors of a number, generate a vector of 10 random integers between -50 and 50, print numbers 1–100 with FizzBuzz logic.

Answer)

(a): Factors of a Number

Code:

```
n <- as.integer(readline("Enter a positive integer: "))
if (is.na(n) || n <= 0) {
  cat("Please enter a positive integer.\n")
} else {
  cat("Factors of", n, "are:\n")
  for (i in 1:n) {
    if (n %% i == 0) {
      cat(i, " ")
    }
  }
  cat("\n")
}
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-7(a) Factors of a Number.R")
Enter a positive integer: 54
Factors of 54 are:
1  2  3  6  9  18  27  54
> |
```

(b): Generate 10 Random Integers Between -50 and 50

Code:

```
set.seed(10)

numbers <- sample(-50:50, 10, replace = TRUE)

print("Random Integers:")

print(numbers)
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/Programs/Experiment-7(b) Generate 10 Random Integers Between -50 and 50.R")
[1] "Random Integers:"
 [1] -42  23  25   4  21   3 -12  32  37 -36
> |
```

(c): FizzBuzz from 1 to 100

Code:

```
for (i in 1:100) {
  if (i %% 3 == 0 && i %% 5 == 0) {
    cat("FizzBuzz\n")
  } else if (i %% 3 == 0) {
    cat("Fizz\n")
  } else if (i %% 5 == 0) {
    cat("Buzz\n")
  } else {
    cat(i, "\n")
  }
}
```

Output:

```
> source("D:/WORD DOCUMENTS/R Programming/Lab/
Programs/Experiment-7(c) FizzBuzz from 1 to 10
0.R")
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
41
Fizz
43
44
FizzBuzz
46
47
Fizz
49
Buzz
Fizz
52
53
Fizz
Buzz
56
Fizz
58
59
FizzBuzz
61
62
Fizz
64
Buzz
Fizz
67
68
Fizz
Buzz
71
Fizz
73
74
FizzBuzz
76
77
```

```
Fizz
79
Buzz
Fizz
82
83
Fizz
Buzz
86
Fizz
88
89
FizzBuzz
91
92
Fizz
94
Buzz
Fizz
97
98
Fizz
Buzz
> 
```